← Back **Plotting functions**
Practice Assignment • 30 min

Due Jul 6, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your latest score.

Next item →

1. Which one of the following graphs represents the function $\ln(-x)$? 1 / 1 point

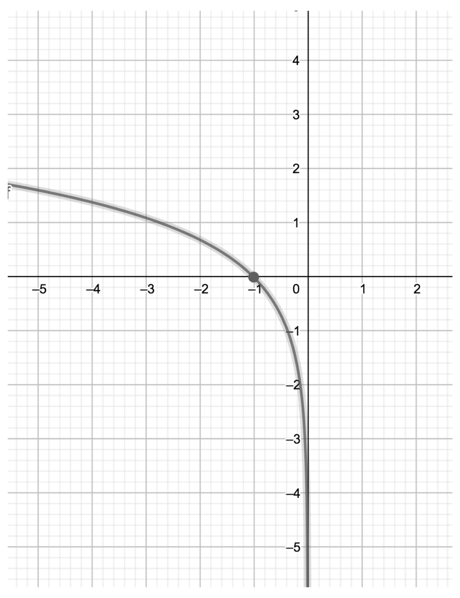

Image description: The domain of this function is $\mathbb{R}^-$ and its corresponding curve is decreasing from f $+\infty$ to $-\infty$ and crosses the x-axis on $x = -1$.

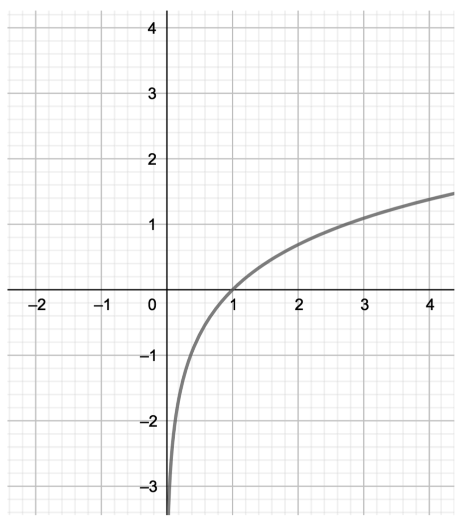

Image description: The domain of this function is $\mathbb{R}^+$ and its corresponding curve isincreasing from $-\infty$ to $+\infty$ and crosses the x-axis on $x = 1$.

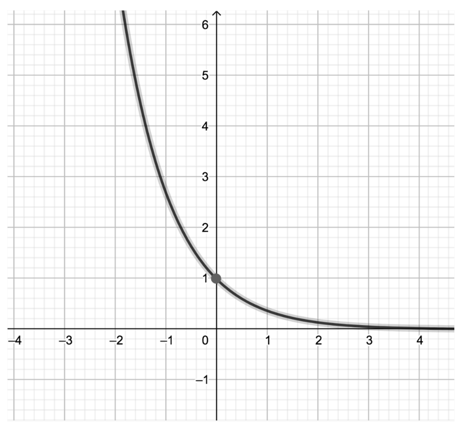

Image description: The domain of this function is $\mathbb{R}$ and its corresponding curve isdecreasing from $+\infty$ to $0$and crosses the y-axis on $y = 1$.

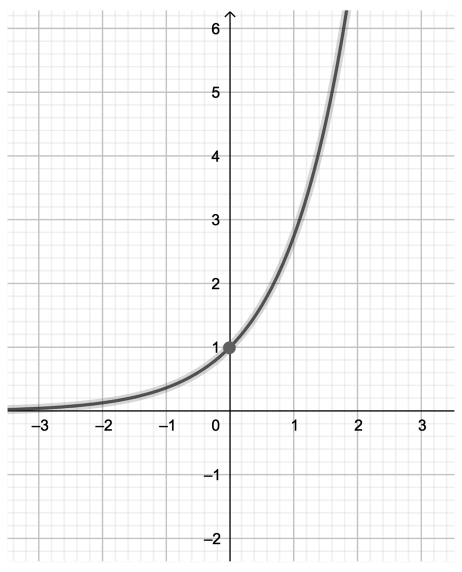

Image description: The domain of this function is $\mathbb{R}$ and its correspondingcurve is increasing from 0 to $+\infty$ and crosses the y-axis on $y = 1$.

✓ Nice work

$\ln(-x)$ is an symmetric $\ln(x)$ with respect tothe y-axis.

2. In the diagram below are three curves labelled $\mathrm{F(x), H(x)}$ and $\mathrm{G(x)}$. 1 / 1 point

Which label represents the curve of the function $p(x) = \ln(x-1)$?

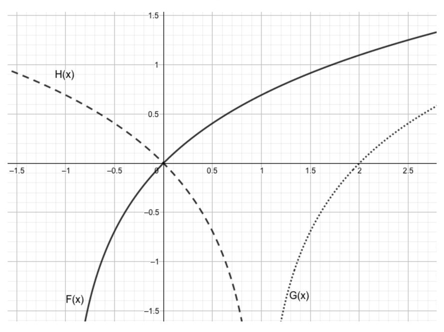

Image description:

The domain for the function H is $(-\infty, 1)$ and its corresponding curve is decreasing from $+\infty$ to $-\infty$ and crosses the x-axis on x = 0.

The domain for the function F is $(-1, +\infty)$ and its corresponding curve is increasing from $-\infty$ to $+\infty$ and crosses the x-axis on $x = 0$.

The domain for the function G is $(1, +\infty)$ and its corresponding curve is increasing from $-\infty$ to $+\infty$ and crosses the x-axis on $x = 2$.

- $\mathrm{F(x)}$
- $\mathrm{H(x)}$
- $\mathrm{G(x)}$

✓ Nice work

Domain of $\ln(x-1) = (1, +\infty)$. The curve for $\ln(x-1)$ crosses the x-axis on point (2, 0) as $\ln(2-1) = ln1 = 0$.

3. Given a function $f(x) = b^x$ where $b \in \mathbb{R}$, which of the following is true if the function f represents an exponential decay? 1 / 1 point

- $b < -1$
- $b > 1$
- $0 < b < 1$
- None of the above.

✓ Nice work

if $0 < b < 1$ the function b^x is a decreasing function and represents an exponential decay

4. The following graph shows the curves of four functions: $2x^2$, $\frac{1}{2}x^2$, $\frac{1}{4}x^2$ and $\frac{1}{8}x^2$. 1 / 1 point

Which label represents the curve of the function $\frac{1}{8}x^2$?

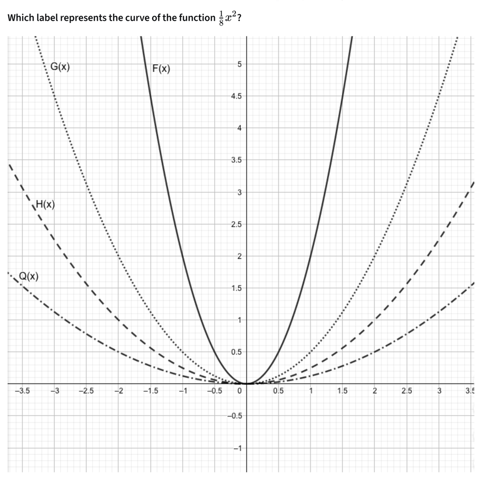

Image description:

The domain all these four functions is $\mathbb{R}$ and their corresponding curves are symmetric with respect to the y-axis.

The curve of function$F$is above that of the function G, H and Q.

The curve of function Q is below that of the functions F, H and G.

The curve of the function H is between the curves of Q and G.

Finally, the G is between the curves of H and F.

- $\mathrm{G(x)}$
- $\mathrm{Q(x)}$
- $\mathrm{F(x)}$
- $\mathrm{H(x)}$

✓ Nice work

$2x^2 > \frac{1}{2}x^2 > \frac{1}{4}x^2 > \frac{1}{8}x^2$. Hence, the curve labelled $\mathrm{Q(x)}$ represents the function $\frac{1}{8}x^2$